Curriculum Vitae

# IVAN B. PENKOV

**Personal** — US citizen, born May 19, 1957, in Sofia, Bulgaria, wife Irina Sousokolova, son Boyan and daughter Anna

## University Education

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| June 1982 | Master Degree in Mathematics<br>Moscow State University<br>Thesis: The inverse Penrose transform<br>Adviser: Professor Yu. I. Manin |
| Oct. 1987 | Candidate of Physical and Mathematical Sciences (Ph.D.),<br>Steklov Mathematical Institute<br>Thesis: Typical modules over classical Lie superalgebras and their geometric realizations<br>Adviser: Professor Yu. I. Manin<br>Official referees: Professors D. Zhelobenko and A. Rudakov |

## Professional Experience

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| 1982–83 | Research Assistant, Institute of Mathematics<br>Bulgarian Academy of Sciences, Sofia |
| May–June 1986 | Visiting Professor at Institute Fourier, Grenoble, France |
| 1987–89 | Research Fellow, Section of Algebra, Institute of Mathematics,<br>Bulgarian Academy of Sciences, Sofia, Bulgaria |
| Oct. 1989 | Visiting Professor at the University of Paris VII, France |
| Jan.–Dec. 1990 | Visiting Assistant Professor, Department of Mathematics,<br>University of California at Berkeley |
| June–July 1990 | Visiting Scholar at the University of Göttingen, Germany |
| Jan.–June 1991 | Visiting Assistant Professor, Department of Mathematics,<br>University of California at Davis |
| July 1991–June 1993 | Assistant Professor, Department of Mathematics,<br>University of California at Riverside |
| Sept.–Oct. 1993 | Visiting Scholar at the Erwin Schrödinger Institute,<br>Vienna, Austria |
| July 1993–June 2000 | Associate Professor (tenured appointment), Department of Mathematics, University of California at Riverside |

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| Apr.–May 1996 | Visiting Scholar and Program Organizer at The Erwin Schrödinger Institute, Vienna, Austria |
| June 1996 | Visiting Scholar at the Max-Planck Institute of Mathematics, Bonn, Germany |
| June 1997 | Visiting Professor at the University of Burgundy Dijon, France |
| Sept. 1997 | Visiting Professor at the University Louis Pasteur, Strasbourg, France |
| Aug.–Sept. 1998 | Visiting Scholar at the Max-Planck Institute of Mathematics, Bonn, Germany |
| Nov. 1999 | Visiting Professor at the University of Hamburg, Germany |
| April-June 2000 | Visiting Scholar at the Max-Planck Institute of Mathematics, Bonn, Germany |
| July 2000 – Oct 2004 | Professor (tenured appointment), Department of Mathematics, University of California at Riverside |
| Aug.–Sept. 2001 | Visiting Scholar at the Max-Planck Institute of Mathematics, Bonn, Germany |
| Jan.–March 2002 | Program organizer and MSRI scholar, MSRI, Berkeley |
| Aug.–Sept. 2002 | Visiting Scholar at the Max-Planck Institute of Mathematics, Bonn, Germany |
| Fall 2003 | Visiting Professor at Yale University |
| Spring 2004 | Visiting Scholar at the Max-Planck Institute of Mathematics, Bonn, Germany |
| Fall 2007 | Visiting Professor at University of California at Berkeley |
| Summer 2008 | Visiting Professor at the University of Sao Paulo, Brazil |
| Spring 2012 | Visiting Professor at Yale University |
| Fall 2012 | Visiting Scholar at the Max-Planck Institute of Mathematics, Bonn, Germany |
| Nov.-Dec. 2018 | Visiting Scholar at the Weizmann Institute of Science, Rehovot, Israel |
| Nov. 2019 | Visiting Professor at the University of Padua, Italy |

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| Fall 2004–Spring 2023 | Professor, Jacobs University Bremen, Germany (named International University Bremen before 2007) |
| Spring 2023–present | Adjunct Professor, Constructor University Bremen, Germany (named Jacobs University Bremen before 2023) |

**Grants/Awards**

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| 1991–1995 | NSF research grant (jointly with R. E. Block and V. Chari). |
| 1995 | NSF grant for organizing the conference in honor of R. E. Block (jointly with V. Chari). |
| 1995 | Grant from the Erwin Schrödinger Institute for the program "Representation Theory with Applications to Mathematical Physics" (jointly with Joseph A. Wolf). |
| 1996 | NSF grant for support of US graduate students at the Erwin Schrödinger Institute. |
| 1995–1997 | NSF research grant. |
| 1997–2002 | NSF Group Infrastructure Grant "Lie Groups, Lie Algebras and Their Representations" (jointly with Joseph A. Wolf and Geoffrey Mason). |
| Jan–May, 2002 | Coorganizer of the one-semester MSRI Program "Representations of Infinite Dimensional Algebras and Mathematical Physics" (jointly with Edward Frenkel, Victor Kac, Vera Serganova and Gregg Zuckerman). |
| April 2002 | Coorganizer of the MSRI Workshop "Quantum field theory and supersymmetry". |
| Sept. 2003 | Coorganizer of the workshop "Locally finite Lie algebras at the Banff Research Institute, Canada. |
| 2003–2006 | NSF Grant "Lie Groups, Lie Algebras and Their Representations" (jointly with Joseph A. Wolf and Geoffrey Mason). |
| 2009–2012 | DFG grant (one postdoc position). |
| 2009 | Volkswagen Foundation grant for organizing the summer school "Structures in Lie Representation Theory" (jointly with A. Joseph and A. Melnikov). |
| 2009-2012 | DFG grant within the Schwerpunktprogramm 1388 "Darstellungstheorie". |

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| 2012-2013 | DFG grant for German-Israeli cooperation. |
| 2012-2015 | DFG grant within the Schwerpunktprogramm 1388 "Darstellungstheorie". |
| 2014 | Coorganizer of the annual meeting in Soltau of the priority program "SPP 1388 Darstellungstheorie". |
| 2014 | Coorganizer of the meeting "Hodge theory and unitary representations of reductive Lie groups", Jacobs University Bremen. |
| 2015 | Coorganizer of the meeting "Groups and Rings Theory and Applications", July 15-22, Sofia, Bulgaria. |
| 2015-2016 | AIM SQuaRE grant |
| 2015–2017 | DFG grant (3/4 postdoc position) |
| 2017 | Vera Serganova and Joseph A. Wolf organized a two-day workshop on the occasion of my 60th birthday at UC Berkeley on May 13-14. |
| 2018-2020 | DFG grant (one postdoc position) |
| 2019 | Coorganizer of the meeting "Perspectives on Representations of Semi-Simple Lie algebras" in honor of Gregg Zuckerman on the occasion of his 70-th birthday. Coorganizer of the mini-conference in honor of Alan Huckleberry at Jacobs University. |
| 2020-2023 | DFG grant (one postdoc position) |
| 2023 | Batsheva de Rothschild Fellow, Israel |
| 2023-2025 | DFG grant (3/4 postdoc position) |

**Main Research Interests**

Representations of finite and infinite dimensional Lie algebras, generalized Harish-Chandra modules, representations of Lie superalgebras and Lie supergroups, geometry of supermanifolds, homogeneous superspaces, super $D$-modules, homogeneous ind-spaces and geometry of ind-varieties.

**Other Professional Activities**

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| 1991–2006 | Founding coorganizer of the California Lie Theory Program, a cooperation program in Lie Theory between the campuses of the University of California. |

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| 2004 | Founding coorganizer of the Program "Darstellungstheorie, Transformationsgruppen und Mathematische Physik" of the consortium of five universities: IU-Bremen and the Universities of Bochum, Darmstadt, Hamburg, and Köln. |
| 2005 | Coorganizer of the conference in honor of Professor Yu. I. Manin at the Max Planck Institute for Mathematics, February 2005.<br>Coorganizer of the International Conference in Honor of R.O. Wells, Jr., IUB, December 1-2, 2005. |
| 2010 | Coorganizer of the International Conference in "Infinite-dimensional Lie Algebras" in Newfoundland, Canada. |
| 2011 | Coorganizer of the conference "Lie Groups" in honor of J.A. Wolf, Bochum, Germany. |
| 2012 | Coorganizer of the German-Israeli workshop "Symplectic Geometry with applications to Representation Theory" at Jacobs University, Bremen, Germany.<br>Organizer of the meeting "7 1/2" in honor of Yu. I. Manin at IHES, Bures sur Yvette, France. |
| 2013 | Organzier of the meeting "Geometry" in honor of Vasil Tsanov, Sofia, Bulgaria. |
| 2014 | Coorganizer of the annual meeting in Soltau of the priority program "SPP 1388 Darstellungstheorie".<br>Coorganizer of the meeting "Hodge theory and unitary representations of reductive Lie groups", Jacobs University Bremen. |
| 2019 | Coorganizer of the meeting "Perspectives on Representations of Semi-Simple Lie algebras" in honor of Gregg Zuckerman on the occasion of his 70-th birthday. |

List of Publications
**IVAN B. PENKOV**

**Research papers**

1. The Penrose transform on general grassmannians, C.R. Acad. Sci. Bulg., 33 (1980), 1439-1442.

2. Null-geodesics of complex Einstein spaces, Funk. Anal. i Priloz., 16, No.1 (1982), 78-79,(Russian), (with Yu.I.Manin).

3. Linear differential operators and cohomology of analytic spaces, Uspekhi Mat. Nauk, 37, No 4 (1982), 171-172, (Russian).

4. D-modules on supermanifolds, Invent.Math., 71 (1983), 501-512.

5. The formalism of left and right connections on supermanifolds, (1982), Lect. on supermanifolds, geom. methods and conf. groups, editor: Döbner et al, World Scientific, Singapore, 1989, 3-13 (with Yu. I. Manin).

6. An introduction to geometric representation theory for complex simple Lie superalgebras, Proc. Conf. Diff. Geom. Math. in Physics, Shumen 1984, World Scientific 1986, 89-106.

7. Serre duality for projective supermanifolds, Funk. Anal. i Priloz., 18,No.1 (1984), 78-79, (Russian) (with O. Ogievetsky).

8. Projectivity and D-affinity of flag supermanifolds, Uspekhi Math. Nauk, 40 No.1 (1985), 211-212, (Russian) (with I. Skornyakov).

9. A geometric approach to the linear Penrose transform, Transactions AMS, 290 (1985), 555-575.

10. Cohomologie des D-modules typiques sur les supervariétés de drapeaux, C. R.-Acad. Sci., Paris, 199 (1984), 1005-1008 (with I. Skornyakov).

11. Characters of typical irreducible finite dimensional $q(m)$-modules, Funk. Anal. i Priloz., 20 (1986), No. 1, 37-45 (Russian).

12. Localisation des representations typiques d'une superalgèbre de Lie complexe classique, C. R. Acad. Sci. Paris, 304 (1987), 163-166.

13. Borel-Weil-Bott theory for classical Lie supergroups, Curr. Probl. Math., Moscow, 32 (1988), 71-124 (Russian).

14. Elements of supergeometry, Curr. Probl. Math., Moscow, 32 (1988), 3-25, (Russian), (with Yu. I. Manin and A. Voronov).

15. Character formulas for some classes of atypical $gl(m + n\varepsilon)$-and $p(m)$-modules, Lett. Math. Phys., 16 (1988), 251-261 (with V. Serganova).

16. Cohomology of $G/P$ for classical complex Lie supergroups $G$ and characters of some atypical G-modules, Annales Fourier, 39 (1989), 845-873, (with V. Serganova).

17. Generic irreducible representations of classical Lie superalgebras, Lecture Notes in Physics, 375 (1991), Springer Verlag, Berlin, 311-319 (with V. Serganova).

18. Representations of classical Lie superalgebras of type 1, Indag. Math., N.S. 3(2), (1992), 419-466 (with V. Serganova).

19. Generic representations of classical Lie superalgebras and their localisation, Monatsh. Math., 118 (1994), 267-313.

20. Generic irreducible representations of finite-dimensional Lie superalgebras, Intern. Journ. Math., 5 (1994), 389-419 (with V. Serganova).

21. Partially and fully integrable modules over Lie superalgebras, Studies in Advanced Mathematics (series editor S.-T. Yau), 4, AMS and Internatl. Press 1997, 49-67 (with I. Dimitrov).

22. Characters of irreducible $G$-modules and cohomology of $G/P$ for the Lie supergroup $G = Q(n)$, Journ. of Math. Sci. 84 (1997), 1382-1412 (with V. Serganova).

23. Characters of finite-dimensional irreducible $q(n)$-modules, Lett. Math. Phys. 40 (1997), 147-158 (with V. Serganova).

24. Characters of strongly generic irreducible Lie superalgebra representations, Intern. Journ. Math. 9 (1998), 331-366.

25. Partially integrable highest weight modules, Transformation Groups 3 (1998), 241-253 (with I. Dimitrov).

26. The support of an irreducible Lie algebra representation, Journ. of Algebra 209 (1998), 129-142 (with V. Serganova).

27. Weight modules of direct limit Lie algebras, IMRN 1999, no. 5, 223-249 (with I. Dimitrov).

28. Weight representations of the Cartan type Lie algebras $W_n$ and $\bar{S}_n$, Math. Research Lett. 6 (1999), 397-416. (with V. Serganova).

29. On the structure of weight modules, Transactions AMS 352 (2000), pp. 2857-2869 (with I. Dimitrov and O. Mathieu).

30. Generalized Harish-Chandra modules, Moscow Math. Journal 2 (2002), 753-767 (with V. Serganova).

31. A Bott-Borel-Weil theory for direct limits of algebraic groups, American Journ. of Math. 124 (2002), 955-998 (with I. Dimitrov and J. A. Wolf).

32. Locally finite Lie algebras with root decomposition, Archiv der Math. 80 (2003) 478-485 (with H. Strade).

33. Cartan subalgebras of $gl_\infty$, Canadian Math. Bull., 46 (4), 2003, 597-616 (with K.-H. Neeb).

34. Finite rank vector bundles on inductive limits of grassmannians, IMRN 2003, no. 34, 1871-1887 (with J. Donin).

35. Classically semisimple locally finite Lie superalgebras, Forum Math. 16 (2004), 431-446.

36. On the existence of (g,k)-modules of finite type, Duke Math. Journal 125 (2004), 329-349 (with V. Serganova and G. Zuckerman).

37. A Reduction Theorem for Highest Weight Modules over Toroidal Lie Algebras, Comm. Math. Physics 250(2004), no. 1, 47-53 (with I. Dimitrov and V. Futorny).

38. Generalized Harish-Chandra modules: a new direction of the structure theory of representations, Acta Applicandae Mathematicae, 81(2004), 311-326(with G. Zuckerman).

39. Ind-varieties of generalized flags as homogeneous spaces for classical ind-groups, IMRN 2004, no. 55, 2935-2953(with I. Dimitrov).

40. Borel subalgebras of $gl(\infty)$, Resenhas IME-USP 6 (2004), 111-119(with I. Dimitrov).

41. Generalized Harish-Chandra modules with generic minimal k-type, Asian Journal of Mathematics 8(2004), 795-812(with G. Zuckerman).

42. A construction of generalized Harish-Chandra modules with arbitrary minimal k-type, Canad. Math. Bull. 50 (2007), 603-609 (with G. Zuckerman).

43. Cartan subalgebras of root-reductive Lie algebras, Journal of Algebra, 308 (2007), 583-611 (with E. Dan-Cohen and N. Snyder).

44. A construction of generalized Harish-Chandra modules for locally reductive Lie Algebras, Transformation Groups 13 (2008), 799-817 (with G. Zuckerman).

45. Locally semisimple and maximal subalgebras of the finitary Lie algebras $gl(\infty)$, $sl(\infty)$, $so(\infty)$, and $sp(\infty)$, Journal of Algebra 322 (2009), 2069-2081 (with I. Dimitrov).

46. Parabolic and Levi subalgebras of finitary Lie algebras, IMRN, International Mathematics Research Notices, Article ID rnp169, doi:10.1093/imrn/rnp169 (with E. Dan-Cohen).

47. Parabolic Subgroups of Real Direct Limit Lie Groups, Contemporary Mathematics 499, 2009, pp. 47-59 (with E. Dan-Cohen and J. A. Wolf).

48. Rank 2 vector bundles on ind-grassmannians, Manin Festschrift,in Progress in Mathematics 270, Birkhaeuser Boston, 2009, pp. 555-572 (with A. S. Tikhomirov).

49. Bounded simple (g; sl(2))-modules for rkg = 2, Journal of Lie Theory 20 (2010), 581-615 (with V. Serganova).

50. Tensor representations of classical locally finite Lie algebras, in Developments and Trends in Infinite-Dimensional Lie Theory, Progress in Mathematics 288, Birkhäuser, 2011, pp. 127-150 (with K. Styrkas).

51. On bounded generalized Harish-Chandra modules, Annales de l'Institut Fourier, 62 (2012), 477-496 (with V. Serganova).

52. Triviality of vector bundles on twisted ind-Grassmannians, Mat. Sbornik 202:1 (2011), 1-39 (with A. S. Tikhomirov).

53. A Bott-Borel-Weil theorem for diagonal ind-groups, Canadian Journal of Mathematics, doi:10.4153/CJM-2011-032-6 (with I. Dimitrov).

54. Categories of integrable $sl(\infty)$-, $o(\infty)$-, $sp(\infty)$-modules, in "Representation Theory and Mathematical Physics", Contemporary Mathematics 557, AMS 2011, pp. 335-357 (with V. Serganova).

55. Levi components of parabolic subalgebras of finitary Lie algebras, in "Representation Theory and Mathematical Physics", Contemporary Mathematics 557, AMS 2011, pp. 129-149 (with E. Dan-Cohen).

56. A Koszul category of representations of finitary Lie algebras, Advances of Mathematics 289 (2016), 250-278 (with Elizabeth Dan-Cohen and Vera Serganova).

57. On the structure of the fundamental series of generalized Harish-Chandra modules, Asian Journal of Mathematics , 16(2012), 489-514 (with G. Zuckerman).

58. On Ideals in the enveloping algebra of a locally simple Lie algebra, Int Math Res Notices (2015) Vol. 2015 5196-5228 first published online June 10, 2014 doi:10.1093/imrn/rnu085 (with Alexey Petukhov).

59. Linear Ind-Grassmannians, Pure and Applied Math. Quarterly 10 (2014), 289-323 (with Alexander Tikhomirov).

60. Algebraic methods in the theory of generalized Harish-Chandra modules, Developments and Retrospectives in Lie Theory: Algebraic Methods, Developments in Mathematics, vol. 38, Springer Verlag, pp. 331-350 (with Gregg Zuckerman).

61. Tensor representations of Mackey Lie algebras and their dense subalgebras, Developments and Retrospectives in Lie Theory: Algebraic Methods, Developments in Mathematics, vol. 38, Springer Verlag, pp. 291-330 (with Vera Serganova).

62. On the Barth-Van de Ven-Tyurin-Sato Theorem, Sbornik: Mathematics, **206**:6 (2015), 814-848 (with Alexander Tikhomirov).

63. A categorification of the Boson-Fermion correspondence via Representation Theory of $sl(\infty)$, Comm. Math. Phys. **341**:3 (2016), pp. 911-931 (with Igor Frenkel and Vera Serganova).

64. Annihilators of highest weight $sl(\infty)$-modules, Transformation Groups 21 (2016), 821-849 (with A. Petukhov)

65. Infinite Kostant cascades and centrally generated primitive ideals of $U(n)$ in types $A_\infty$, $C_\infty$, Journal of Algebra **447**(2016), 109-134 (with Mikhail Ignatyev).

66. Schubert decompositions for ind-varieties of generalized flags, Asian Journal of Mathematics 21 (2017), 599-630 (with L. Fresse).

67. On ideals in $U(sl(\infty))$, $U(o(\infty))$, $U(sp(\infty))$. In: Representation theory - current trends and perspectives, EMS Series of Congress Reports, European Mathematical Society (EMS), 2016, 565-602 (with A. Petukhov).

68. Ordered tensor categories and representations of the Mackey Lie algebra of infinite matrices, Algebras and Representation Theory 22 (2019), 249-279 (with A. Chirvasitu).

69. Real group orbits on flag ind-varieties of $\mathrm{SL}(\infty, \mathbb{C})$, In: Lie Theory and Its Applications in Physics, Springer Proceedings in Mathematics and Statistics, vol. 191 (2016), 111-135 (with M. V. Ignatyev and J. A. Wolf).

70. On the categories of admissible $\big(\mathfrak{g}, \mathrm{sl}(2)\big)$-modules,Transformation Groups 23(2) (2018), 463-489(with V. Serganova and G. Zuckerman).

71. Primitive ideals of $U(\mathfrak{sl}(\infty))$, Bulletin LMS 50 (2018), 443-448 (with A. Petukhov).

72. Ind-varieties of generalized flags: a survey of results, Journal of Mathematical Sciences 248 (2020), 255–302 (with M. Ignatyev).

72'. Ind-varieties of generalized flags: a survey of results. In " Itogi nauki i techniki, series Modern mathematics and its applications", Russian Academy of Sciences 2018, Topic Surveys 147, 3–50, arXiv:1701.08478 (with M. Ignatyev), Russian version.

73. Decomposition of cohomology of vector bundles on homogeneous ind-spaces, C. R. Acad. Sci. Bulg. 70:7 (2017), 907-916 (with E. Hristova).

74. Orbit duality in ind-varieties of maximal generalized flags, Transactions of the Moscow Mathematical Society, 2017, 131-160 (with L. Fresse).

74'. Orbit duality in ind-varieties of maximal generalized flags, Transactions of the Moscow Mathematical Society 78 (2017), no 1(with L. Fresse), Russian version, arXiv:1704.03671.

75. Representation categories of Mackey Lie algebras as universal monoidal categories, Pure and Applied Mathematics Quarterly 13 (2017), 77-121 (with A. Chirvasitu).

76. Primitive ideals of $U(\mathfrak{sl}(\infty))$ and the Robinson-Schensted algorithm at infinity, In: Representation of Lie Algebraic Systems and Nilpotent orbits, Progress in Mathematics 330 (2019), Birkhäuser, 471-499 (with A. Petukhov).

77. Integrable $\mathfrak{sl}(\infty)$-modules and category $\mathcal{O}$ for $\mathfrak{gl}(m|n)$, Journal LMS **99** (2019), 403-427 (with C. Hoyt and V. Serganova).

78. On an infinite limit of BGG categories $\mathcal{O}$, Moscow Mathematical Journal 19 (2019), 655-693 (with K. Coulembier).

79. On the existence of infinite-dimensional generalized Harish-Chandra modules, Sao-Paulo Journ. Math. 12(2018), 290-294 (with G. Zuckerman).

80. Simple bounded weight modules of $\mathfrak{sl}(\infty)$, $\mathfrak{o}(\infty)$, $\mathfrak{sp}(\infty)$, Transformation Groups, 25(4) (2020), 1125-1160 (with D. Grantcharov).

81. Large annihilator category $\mathcal{O}$ for $\mathfrak{sl}(\infty)$, $\mathfrak{o}(\infty)$, $\mathfrak{sp}(\infty)$, Journal of Algebra 532 (2019), 249-279 (with V. Serganova).

82. Examples of automorphism groups of ind-varieties of generalized flags, Journal of Geometry and Symmetry in Physics 50 (2018), 71-77.

83. An algebraic-geometric construction of ind-varieties of generalized flags, Annali di Matematica Pura ed Applicata 201 (2022), 2287–2314 (with A. Tikhomirov).

84. Multiple flag ind-varieties with finitely many orbits, Transformation groups 27 (2022), 833–865 (with L. Fresse).

85. Integrable bounded weight modules of classical Lie superalgebras at infinity, Journal of Pure and Applied Algebra 226 (2022), 106847 (with L. Calixto).

86. Universal tensor categories generated by dual pairs, Applied Categorical Structures 29 (5), 915-950 (with A. Chirvasitu).

87. Automorphism groups of ind-varieties of generalized flags, Transformation groups (2022),
doi:10.1007/s00031-022-09703-1 (with M. Ignatyev).

88. Bounded weight modules for basic classical Lie superalgebras at infinity, arXiv:2205.07138 (with D. Grantcharov and V. Serganova).

89. Topological tensor representations of $\mathfrak{gl}(V)$ for a space $V$ of countable dimension, Algebraic Geometry and Physics, to appear, arXiv:2206.00654 (with F. Esposito).

90. Topological semiinfinite tensor (super)modules, arXiv:2301.08921 (with F. Esposito).

91. On homogeneous spaces for diagonal ind-groups, arXiv:2303.18146 (with L. Fresse).

**Monographs**

1. Classical Lie algebras at infinity, Springer Monographs in Mathematics, 2022, ISBN 978-3-030-89659-1 (with C. Hoyt).

**Books edited**

1. V. Chari, I. B. Penkov - editors. Modular Interfaces: Modular Lie Algebras, Quantum Groups, and Lie Superalgebras. "Studies in Advanced Mathematics", vol. 4, 1997, AMS.

2. A. Joseph, A. Melnikov, I. Penkov - editors. Highlights in Lie Algebraic Methods. "Progress in Mathematics", vol. 295, 2011, Birkhauser.

3. A. Huckleberry, I. Penkov, G. Zuckerman - editors. Lie Groups: Structure, Actions, and Representations. "Progress in Mathematics", vol. 306, 2013, Birkhauser.

4. G. Mason, I. Penkov, J.A. Wolf - editors. Developments and Retrospectives in Lie Theory: Geometric and Analytic Methods, Developments in Mathematics, vol.37, 2014, Springer Verlag.

5. G. Mason, I. Penkov, J.A. Wolf - editors. Develpoments and Retrospectives in Lie Theory: Algebraic Methods, Developments in Mathematics, vol. 38, 2014, Springer Verlag.